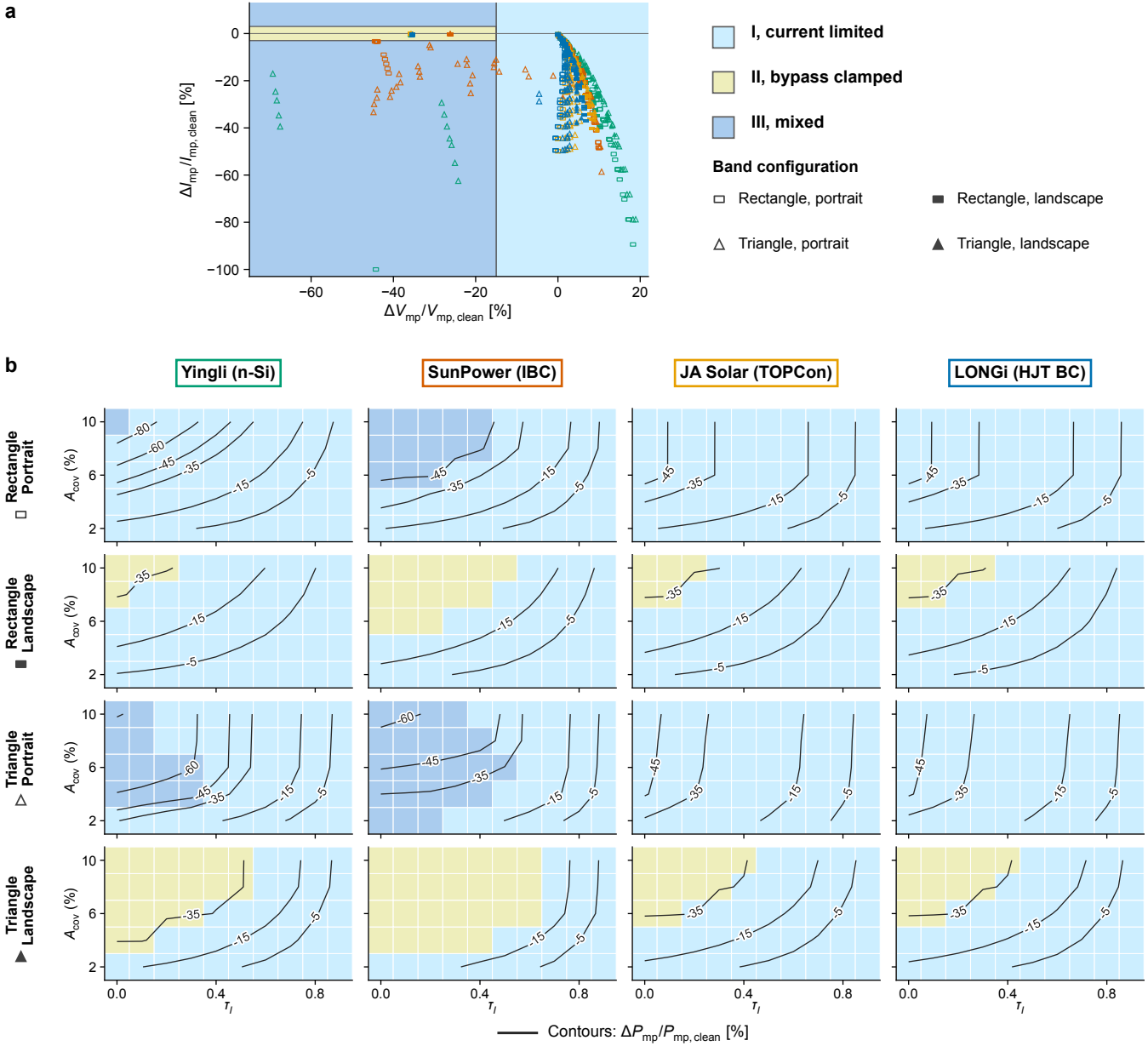
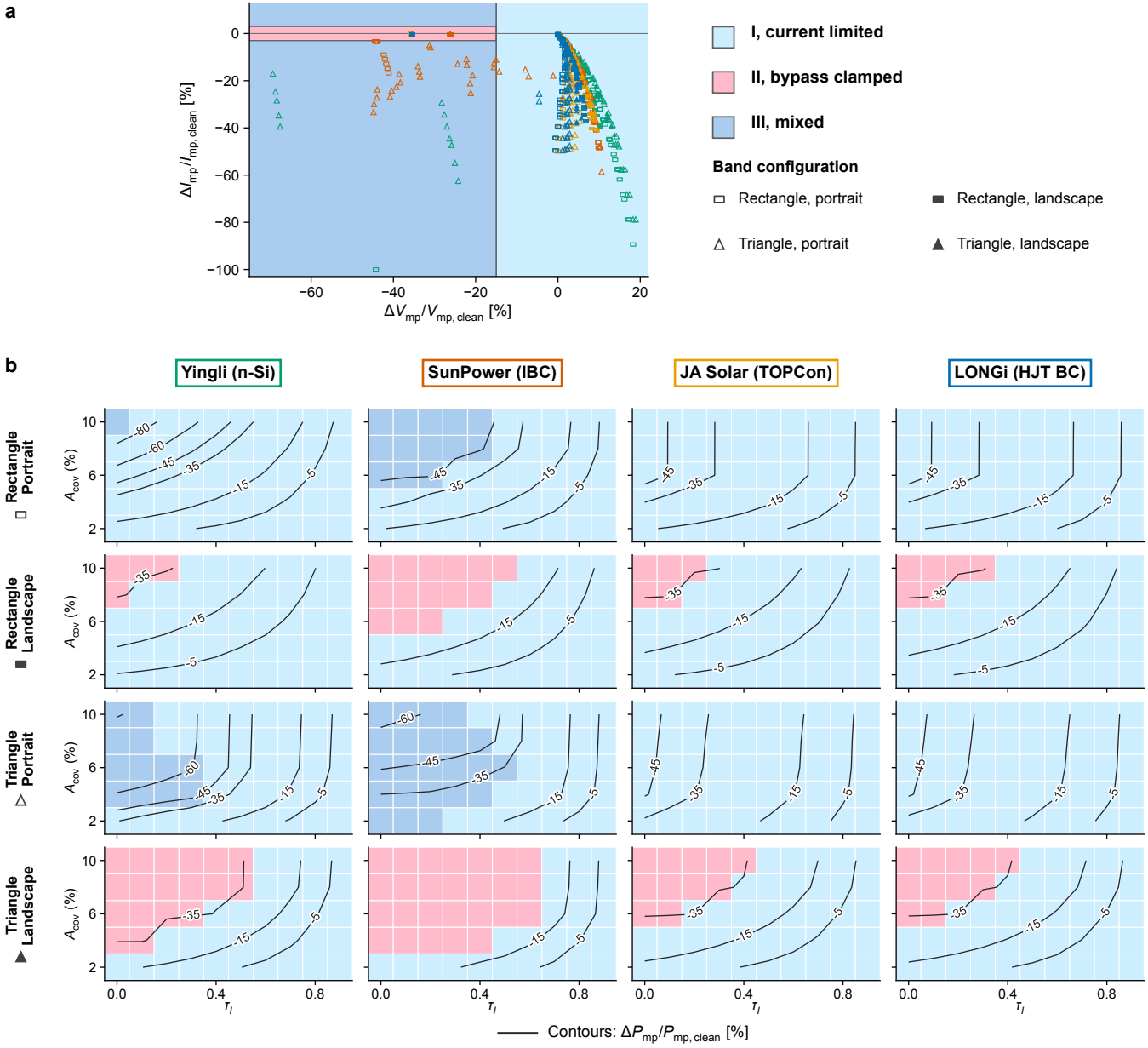
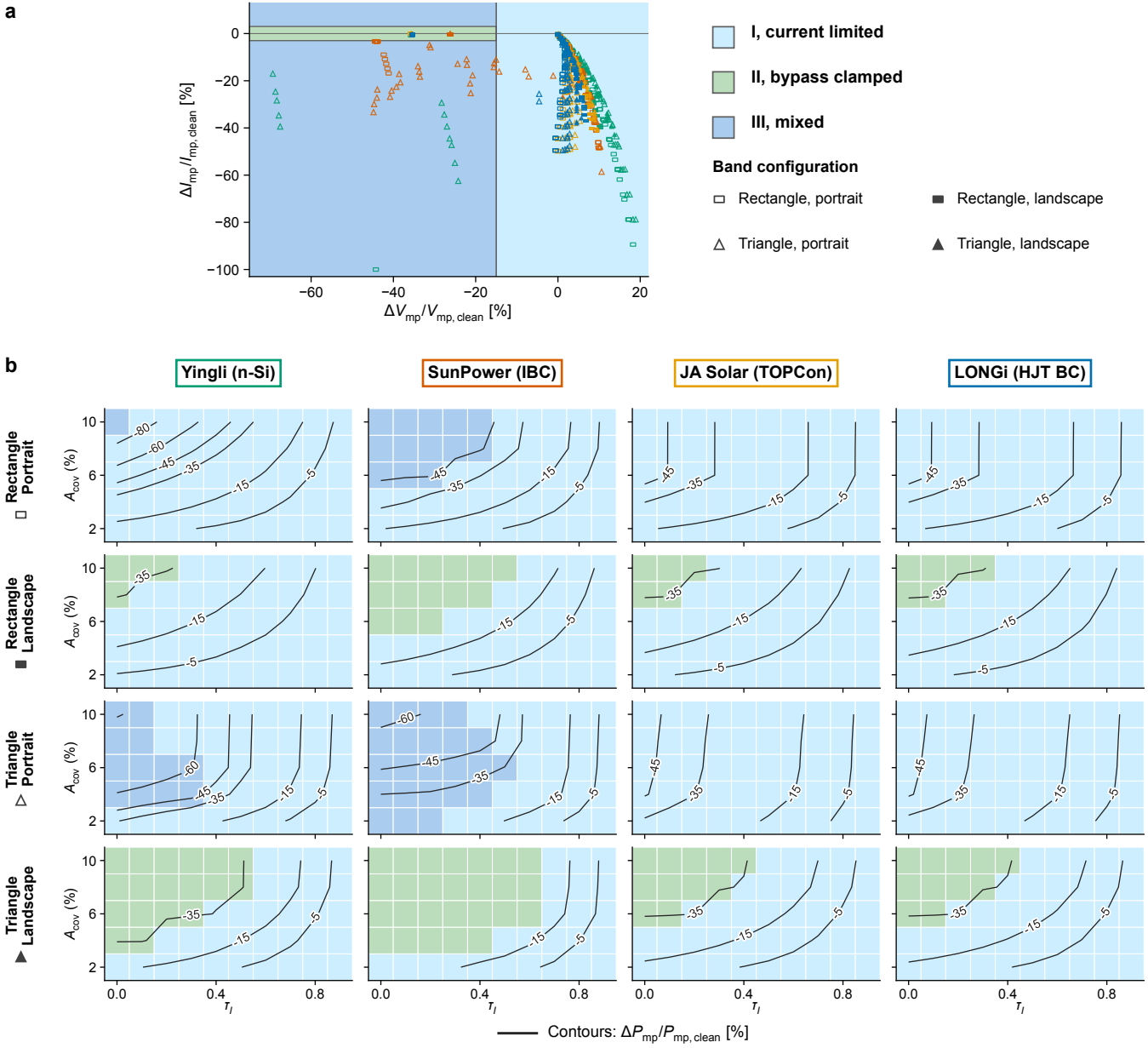
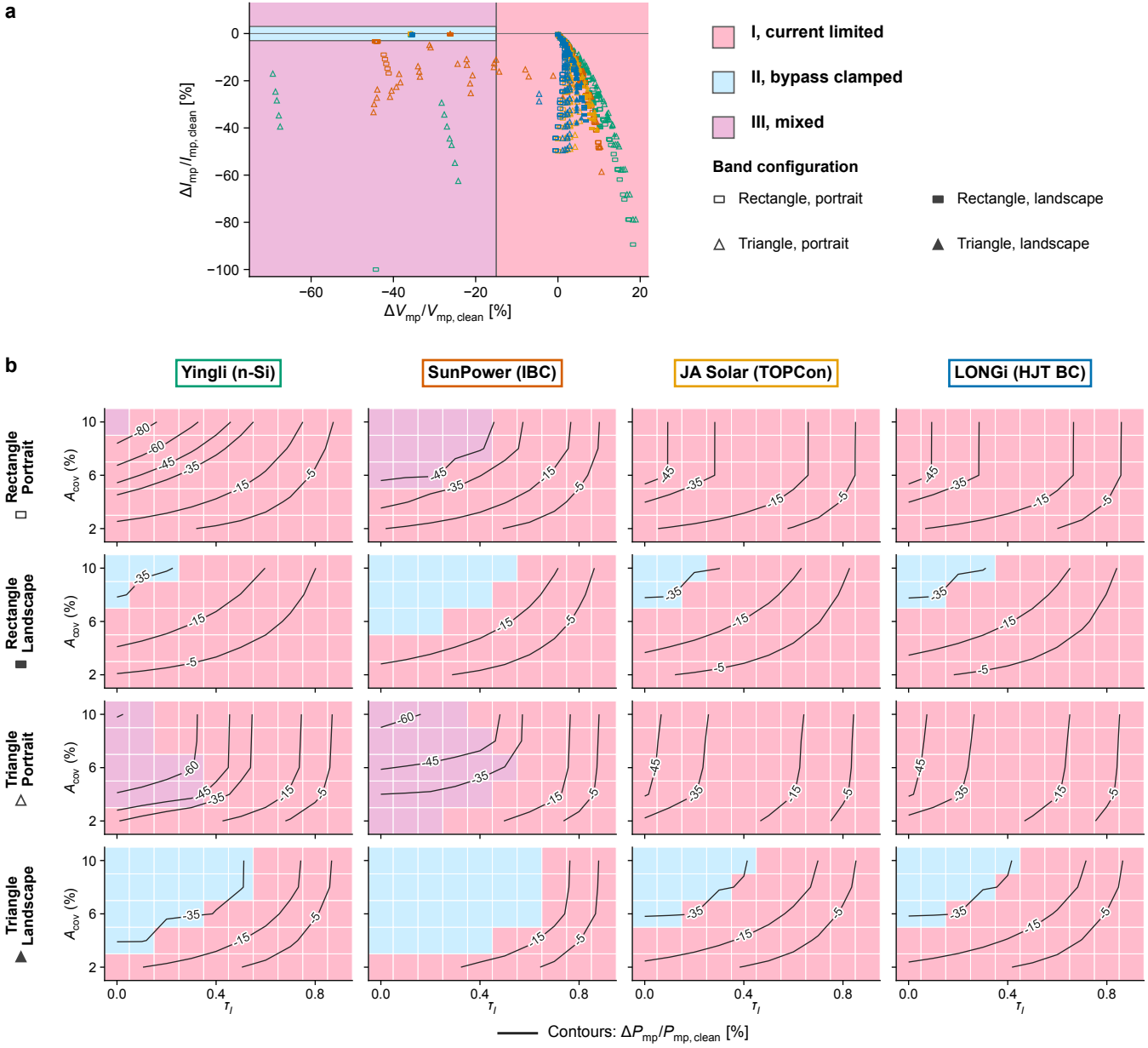
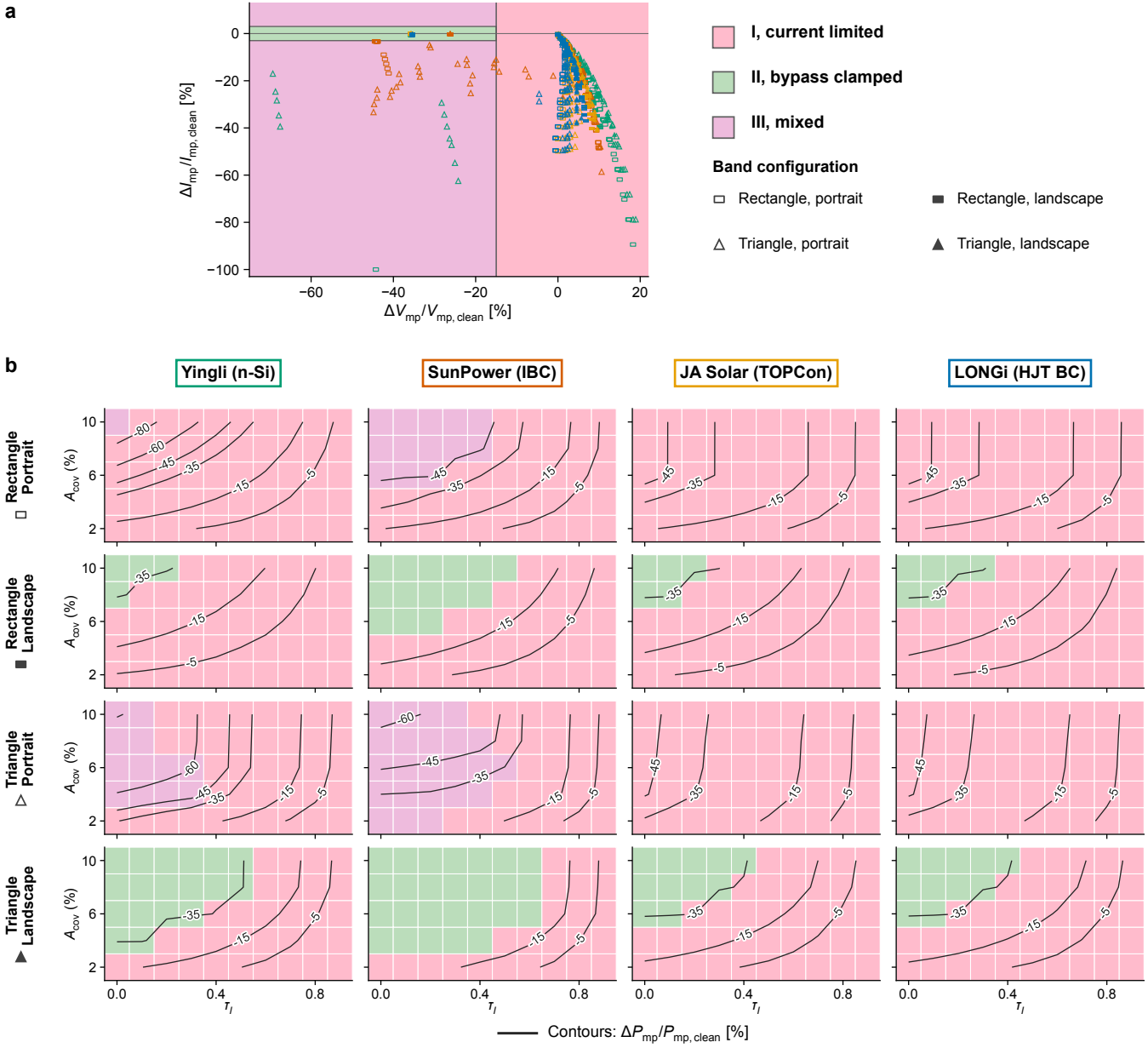
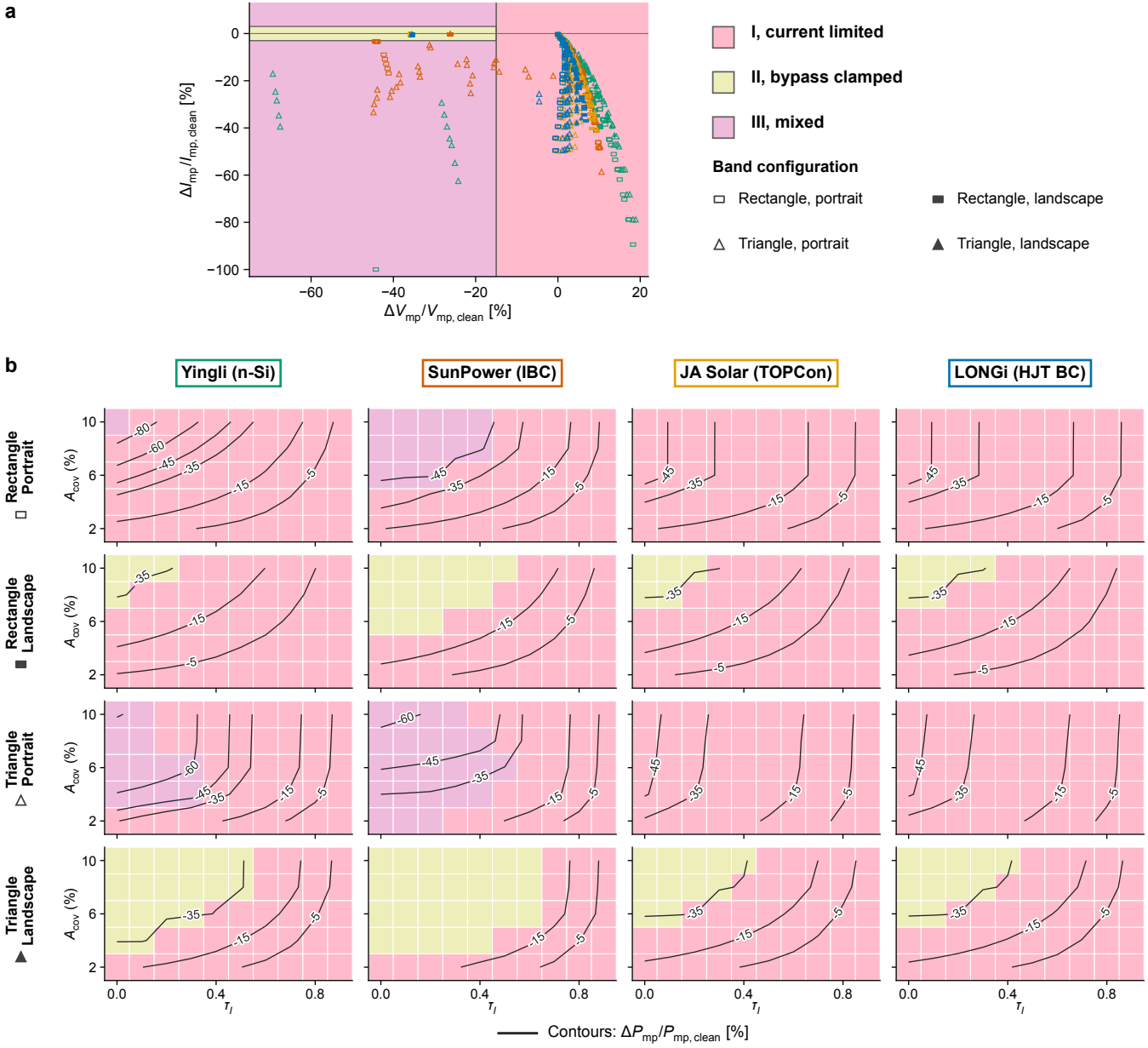

a
I, current limited
II, bypass clamped
III, mixed
Band configuration
Rectangle, portrait
Rectangle, landscape
Triangle, portrait
Triangle, landscape
$\Delta I_{mp}/I_{mp,clean}$ [%]
$\Delta V_{mp}/V_{mp,clean}$ [%]
b
Yingli (n-Si)
SunPower (IBC)
JA Solar (TOPCon)
LONGi (HJT BC)
Rectangle Portrait
Rectangle Landscape
Triangle Portrait
Triangle Landscape
A_{cov} (%)
τ_I
Contours: $\Delta P_{mp}/P_{mp,clean}$ [%]


a
0
−20
−40
−60
−80
−100
$\Delta I_{mp}/I_{mp,clean}$ [%]
−60
−40
−20
0
20
$\Delta V_{mp}/V_{mp,clean}$ [%]
I, current limited
II, bypass clamped
III, mixed
Band configuration
Rectangle, portrait
Rectangle, landscape
Triangle, portrait
Triangle, landscape
b
Yingli (n-Si)
SunPower (IBC)
JA Solar (TOPCon)
LONGi (HJT BC)
Rectangle Portrait
Rectangle Landscape
Triangle Portrait
Triangle Landscape
A_{cov} (%)
10
6
2
0.0
0.4
0.8
τ_I
Contours: $\Delta P_{mp}/P_{mp,clean}$ [%]


a
0
−20
−40
−60
−80
−100
−60
−40
−20
0
20
$\Delta I_{mp}/I_{mp,clean}$ [%]
$\Delta V_{mp}/V_{mp,clean}$ [%]
I, current limited
II, bypass clamped
III, mixed
Band configuration
Rectangle, portrait
Rectangle, landscape
Triangle, portrait
Triangle, landscape
b
Yingli (n-Si)
SunPower (IBC)
JA Solar (TOPCon)
LONGi (HJT BC)
Rectangle Portrait
Rectangle Landscape
Triangle Portrait
Triangle Landscape
A_{cov} (%)
10
6
2
0.0
0.4
0.8
τ_I
-80
-60
-45
-35
-15
-5
Contours: $\Delta P_{mp}/P_{mp,clean}$ [%]


a
0
−20
−40
−60
−80
−100
−60
−40
−20
0
20
ΔImp/Imp, clean [%]
ΔVmp/Vmp, clean [%]
I, current limited
II, bypass clamped
III, mixed
Band configuration
Rectangle, portrait
Rectangle, landscape
Triangle, portrait
Triangle, landscape
b
Yingli (n-Si)
SunPower (IBC)
JA Solar (TOPCon)
LONGi (HJT BC)
Rectangle Portrait
Rectangle Landscape
Triangle Portrait
Triangle Landscape
Acov (%)
τl
Contours: ΔPmp/Pmp, clean [%]


a
0
−20
−40
−60
−80
−100
−60
−40
−20
0
20
$\Delta I_{mp}/I_{mp,clean}$ [%]
$\Delta V_{mp}/V_{mp,clean}$ [%]
I, current limited
II, bypass clamped
III, mixed
Band configuration
Rectangle, portrait
Rectangle, landscape
Triangle, portrait
Triangle, landscape
b
Yingli (n-Si)
SunPower (IBC)
JA Solar (TOPCon)
LONGi (HJT BC)
Rectangle Portrait
Rectangle Landscape
Triangle Portrait
Triangle Landscape
A_{cov} (%)
10
6
2
0.0
0.4
0.8
τ_l
-80
-60
-45
-35
-15
-5
Contours: $\Delta P_{mp}/P_{mp,clean}$ [%]


a
0
−20
−40
−60
−80
−100
−60
−40
−20
0
20
ΔImp/Imp, clean [%]
ΔVmp/Vmp, clean [%]
I, current limited
II, bypass clamped
III, mixed
Band configuration
Rectangle, portrait
Rectangle, landscape
Triangle, portrait
Triangle, landscape
b
Yingli (n-Si)
SunPower (IBC)
JA Solar (TOPCon)
LONGi (HJT BC)
Rectangle Portrait
Rectangle Landscape
Triangle Portrait
Triangle Landscape
Acov (%)
τI
0.0
0.4
0.8
Contours: ΔPmp/Pmp, clean [%]